

A Fairness-Aware Constraint Satisfaction Framework for University Exam Timetabling with Dual-Role Instructor Conflict Modeling

1st Jibrailu Daniel*

Artificial Intelligence Engineering
Computer Engineering Dept., Istanbul Okan University
Istanbul, Türkiye
jdaniel@stu.okan.edu.tr

3rd Emin Çağan Apaydın

Computer Engineering
Computer Engineering Dept., Istanbul Okan University
Istanbul, Türkiye
emapaydin@stu.okan.edu.tr

5th Mutaz Wael

Computer Engineering
Computer Engineering Dept., Istanbul Okan University
Istanbul, Türkiye
muabdalгани@stu.okan.edu.tr

7th Sudenur Yanık

Computer Engineering
Computer Engineering Dept., Istanbul Okan University
Istanbul, Türkiye
sudyanik@stu.okan.edu.tr

2nd Yusuf Garga Bello

Artificial Intelligence Engineering
Computer Engineering Dept., Istanbul Okan University
Istanbul, Türkiye
ybello@stu.okan.edu.tr

4th Atakan Onay

Computer Engineering
Computer Engineering Dept., Istanbul Okan University
Istanbul, Türkiye
atonay@stu.okan.edu.tr

6th Nevşin Nağas

Computer Engineering
Computer Engineering Dept., Istanbul Okan University
Istanbul, Türkiye
nnagas@stu.okan.edu.tr

Abstract—Exam scheduling at universities is an NP-hard problem due to high levels of constraint interaction. Existing Constraint Programming (CP) approaches to the University Exam Timetabling Problem (UETP) treat the instructor pool and the student body as mutually exclusive sets, omit room assignment from the constraint model, and address workload balance only as a soft post-processing heuristic rather than as a first-class optimisation objective. This paper closes all three gaps within a single, unified Constraint Satisfaction Optimisation Problem (CSOP) framework. The three principal novelties over prior CP-based work are: (i) a formally encoded dual-role instructor conflict hard constraint (H4) that prevents PhD candidates from being simultaneously examined and assigned as invigilators, a constraint class absent from all reviewed CP formulations; (ii) a multi-room Boolean assignment matrix that allows large-enrollment examinations to span multiple physical venues within the CP model itself, rather than being resolved by a post-processing heuristic; and (iii) a min-max workload fairness proxy integrated as a primary weighted CP-SAT objective, providing a provably tighter per-instructor bound than the variance minimisation objectives used in prior work. The model is solved by the Google OR-Tools CP-SAT solver. Experiments on four instances of the Carter/Toronto benchmark suite and on real institutional data from a metropolitan university demonstrate that all hard constraints are satisfied across every tested instance, near-optimal workload fairness (min-max gap 0–3) is achieved on

three of four benchmarks, and the monolithic formulation scales to 189 examinations within practical time limits. A structural scalability boundary is identified at approximately 300 examinations for the monolithic formulation, beyond which column generation decomposition is required. An ablation study exposes a fundamental tradeoff between consecutive-invigilation avoidance and long-term fairness.

Index Terms—constraint satisfaction, exam timetabling, multi-objective optimisation, CP-SAT, educational administration, workload fairness, constraint programming

I. INTRODUCTION

University Exam Timetabling involves scheduling examinations within strictly defined timeslots and physical rooms such that no operational conflicts occur. This process is a cornerstone of modern educational administration, ensuring optimal use of scarce institutional resources while satisfying the diverse operational, pedagogical, and personal needs of various stakeholders. The University Exam Timetabling Problem (UETP), a prominent subset of this domain, is mathematically classified as NP-hard, indicating that no known algorithm can solve all possible instances in polynomial time.

The primary objective of UETP is to assign a discrete set of examinations to a limited set of timeslots and physical rooms. This assignment must strictly adhere to inviolable hard constraints such as preventing overlapping exams for any enrolled student and respecting the physical seating capacities of rooms, while simultaneously navigating a multi-objective landscape of soft constraints that minimise penalties associated with stakeholder preferences, consecutive examination fatigue, and workload distribution among faculty and staff.

A. Limitations of Existing CP-Based Approaches

Despite the natural alignment between CP and scheduling feasibility, existing CP and CSOP formulations for UETP share three systematic limitations that motivate the present work.

Gap 1 – Missing dual-role conflicts. All reviewed CP models treat the instructor pool and the student body as disjoint sets. In practice, teaching assistants and PhD candidates are frequently enrolled in graduate-level courses while being simultaneously mandated to invigilate undergraduate examinations. Comlan and Allohombou [7] acknowledge this dual-role phenomenon in the LMD timetabling framework but resolve it through manual administrative rules rather than formal constraint encoding. No existing CP formulation encodes this conflict as a hard constraint inside the solver.

Gap 2 – Room assignment decoupled from the CP model. The Toronto/Carter benchmarks, which underpin virtually all CP comparison tables in the literature, are deliberately uncapacitated: rooms are absent from the problem definition [4]. As a result, CP models evaluated on these benchmarks do not include room-assignment constraints. When room assignment is addressed it is typically handled as a separate post-processing step, preventing the solver from exploiting the interdependence between timeslot and room assignment when large-enrollment examinations require dynamic partitioning across multiple venues.

Gap 3 – Workload fairness as a secondary heuristic. Gültekin et al. [8] include a workload-balancing heuristic for midterm scheduling as a soft post-processing step; Ghaffar et al. [11] confirm in a systematic review that fairness and invigilator modelling are nearly universally absent from reviewed formulations. Where workload objectives do appear they are expressed as variance minimisation, a quadratic term that CP-SAT does not support natively and that provides a weaker per-instructor guarantee than a direct min-max bound.

B. Contributions

This paper addresses all three gaps within a single unified CSOP. The primary contributions are:

- 1) **Hard-constraint dual-role conflict (H4):** A formal CP constraint preventing any instructor from invigilating an examination assigned to the same timeslot as an examination in which they are simultaneously enrolled, to the best of our knowledge the first such encoding in an exact CP solver.

- 2) **Integrated multi-room Boolean assignment:** A Boolean matrix Y_{er} enabling large-enrollment examinations to span multiple physical venues as first-class CP decision variables, allowing the solver to co-optimize timeslot and room assignment jointly.
- 3) **Min-max fairness as a primary CP-SAT objective:** A linear min-max proxy $S_2 = \max_i(\text{load}_i) - \min_i(\text{load}_i)$ replacing unsupported quadratic variance and providing a provably stronger bound against worst-case overload.
- 4) **Comprehensive experimental validation:** Four Carter benchmark instances, a real institutional dataset (131 examinations, 36 instructors including 25 dual-role PhD candidates), and an ablation study quantifying the S_3 consecutive-invigilation tradeoff.

II. LITERATURE REVIEW

A. Classical Timetabling, Graph Colouring, and the Carter Benchmark

The modern examination timetabling literature traces its benchmark foundation to Carter, Laporte, and Lee [1], who introduced the EXAMINE system and 13 real-world problem instances universally known as the Toronto or Carter benchmarks. These instances define the standard evaluation protocol: a proximity cost function penalising conflicting student exams scheduled close together, with weights 16, 8, 4, 2, and 1 for timeslot gaps of 1–5 respectively, framing the problem as graph colouring over a student conflict graph. A key observation, later formalised by Ceschia et al. [4], is that the Toronto benchmark is deliberately uncapacitated: rooms are absent, invigilators are absent, and the sole hard constraint is conflict-freedom. This design choice directly explains why prior CP models evaluated on this benchmark do not incorporate room or invigilator constraints, and directly motivates our choice to model both explicitly.

Burke and Newall [2] demonstrated that adaptive heuristic ordering consistently produces results at or above the state of the art across all 13 Carter instances. Qu et al. [3] subsequently provided the definitive survey of examination timetabling search methodologies, establishing the authoritative comparative tables cited by subsequent work. Studies have further shown that the modularity of constraint networks strongly correlates with algorithm search time [9].

B. Metaheuristics in Educational Scheduling

Genetic algorithms (GA) have been extensively applied to multi-objective exam timetabling, balancing student convenience against resource penalties [10]. Ghaffar et al. [11] conducted a systematic review concluding that hybrid designs combining constructive heuristics with local search consistently outperform pure GA methods, while also noting the near-universal absence of fairness and invigilator modelling in reviewed formulations. Iterated Local Search [19], Tabu Search, and ACO combined with Lévy Flight [18] have been successfully adapted to escape local optima in complex timetabling datasets. Mandal et al. [5] proposed a partial graph heuristic with modified great deluge (PGH-mGD) achieving

best-known or second-best proximity cost results on most Carter instances. Multi-neighbourhood local search with room-split balancing [17] is the nearest prior work to our anti-fragmentation term S_5 , though it operates as a heuristic without hard feasibility guarantees.

C. Constraint Programming and Exact Solvers: Where Prior Work Falls Short

The Constraint Satisfaction Problem (CSP) provides a rigorous mathematical foundation for timetabling. Alrawashdeh et al. [9] demonstrated that profiling-based CSP algorithms achieve feasible schedules with efficient classroom utilisation, validating constraint networks for real institutional data. However, the formulation encodes only student-conflict and capacity constraints; invigilator assignment is entirely absent. Modern exact solvers, particularly those utilising SAT and Lazy Clause Generation (LCG) such as Google OR-Tools CP-SAT, have bridged the gap between theoretical constraint programming and practical performance. Recent work explores LLMs and Graph Neural Networks (GNNs) coupled with classical constraint reasoning to predict propagation and solution validity [12], [20].

Table I systematically maps the constraint features of representative CP-based timetabling approaches, making explicit that no prior CP formulation simultaneously encodes dual-role conflicts, integrates room assignment as a CP decision variable, and optimises workload fairness as a primary objective.

TABLE I
FEATURE COMPARISON OF CP-BASED EXAM TIMETABLING APPROACHES

Method	Room CP	Invig.	Dual-role	Fairness
Alrawashdeh et al. [9]	✓	×	×	×
Comlan & Allohoubou [7]	×	Manual	Manual	×
Gültekin et al. [8]	×	Post	×	Post
Bazari et al. [6]	×	×	Partial	×
Xiaofan et al. [13]	×	×	×	×
This work	✓	✓	✓	✓

Post = post-processing heuristic only.

D. Dual-Role Instructor Conflicts and Workload Fairness

Bazari et al. [6] modelled professor exam-authoring conflicts using MIP, a related but distinct problem, while Comlan and Allohoubou [7] acknowledge dual-role doctoral student conflicts in the LMD framework but resolve them through manual rules rather than formal constraint encoding. To our knowledge, the present work is the first to encode dual-role conflict as a formal CP hard constraint (H4) within an exact solver. Gültekin et al. [8] include a workload balancing heuristic as a soft post-processing step; our min-max proxy (S_2) is the first to integrate fairness as a primary weighted optimisation objective within an exact CP-SAT framework.

E. AI-Augmented Constraint Reasoning

Bonlarron et al. [12] demonstrated at IJCAI-25 that LLMs can be coupled with constraint propagation engines to translate

Algorithm 1 Conflict Graph Construction

Require: Exams $\mathcal{E}$, student enrollments $S(e) \forall e \in \mathcal{E}$

Ensure: Conflict matrix C : $C_{jk} = 1$ iff exams j, k share students

```

1: Initialise  $C_{jk} \leftarrow 0$  for all  $j, k$ 
2: for each pair  $(e_j, e_k)$  with  $j < k$  do
3:   if  $S(e_j) \cap S(e_k) \neq \emptyset$  then
4:      $C_{jk} \leftarrow 1$ 
5:   end if
6: end for
7: return  $C$ 

```

natural-language problem descriptions into formal constraint clauses, directly relevant to real-world deployments. GNNs are being explored as approximate CSP solvers [20], representing a promising warm-starting direction for large instances that exceed the current formulation’s memory capacity.

III. PROBLEM FORMULATION

The UETP is formulated as a CSOP with problem instance $\mathcal{P} = (\mathcal{E}, \mathcal{T}, \mathcal{R}, \mathcal{I})$, where $\mathcal{E}$ is the set of examinations, $\mathcal{T}$ the set of available timeslots, $\mathcal{R}$ the set of rooms (physical and virtual), and $\mathcal{I}$ the set of instructors eligible for invigilation. Each room $r \in \mathcal{R}$ has capacity $\text{cap}(r)$, with one virtual room of capacity 100,000 serving as an unlimited-concurrency sink for online examinations. Each examination $e \in \mathcal{E}$ is annotated with enrolled students $S(e)$, responsible lecturer $\text{lec}(e) \in \mathcal{I}$, and required invigilator count $\text{req}(e)$.

A. Decision Variables

Timeslot Assignment X_e : integer variable for the timeslot assigned to examination e :

$$X_e \in \{0, 1, \dots, |\mathcal{T}| - 1\} \quad \forall e \in \mathcal{E} \quad (1)$$

Room Assignment Y_{er} : Boolean matrix enabling large-enrollment exams to span multiple classrooms simultaneously:

$$Y_{er} \in \{0, 1\} \quad \forall e \in \mathcal{E}, r \in \mathcal{R} \quad (2)$$

Invigilator Assignment Z_{ei} : Boolean variable indicating whether instructor i invigilates examination e :

$$Z_{ei} \in \{0, 1\} \quad \forall e \in \mathcal{E}, i \in \mathcal{I} \quad (3)$$

B. Student Conflict Graph

A student conflict graph $G = (\mathcal{E}, C)$ has edge $(e_1, e_2) \in C$ iff $S(e_1) \cap S(e_2) \neq \emptyset$. Construction runs in $O(|\mathcal{E}|^2 \cdot |\bar{S}|)$ time via set intersection, where $|\bar{S}|$ is the average class size.

IV. MATHEMATICAL CONSTRAINTS METHODOLOGY

A. Inviolable Hard Constraints (H1–H6)

H1 – No Student Time Conflict. For every edge $(e_1, e_2) \in C$:

$$X_{e_1} \neq X_{e_2} \quad (4)$$

H2 – Room Capacity and Online Routing. Aggregate capacity of assigned rooms must meet or exceed enrollment;

online exams route to the virtual room; overflow incurs a 5,000-unit penalty:

$$\sum_{r \in \mathcal{R}} Y_{er} \cdot \text{cap}(r) \geq |S(e)| \quad \forall e \in \mathcal{E} \quad (5)$$

H3 – No Room Clash. Let $\text{ss}_{e_1 e_2}$ be a reified Boolean for $(X_{e_1} = X_{e_2})$:

$$\text{ss}_{e_1 e_2} \Rightarrow Y_{e_1 r} + Y_{e_2 r} \leq 1 \quad \forall r \in \mathcal{R}_{\text{phys}} \quad (6)$$

H4 – Dual-Role Instructor Conflict (Novel Contribution). This constraint is the first formal CP encoding of the dual-role phenomenon. Prior CP approaches either assume instructors cannot be students (rendering the constraint vacuous) or resolve conflicts through post-hoc manual rules [7]. By encoding H4 as a hard clause inside CP-SAT, the solver can propagate dual-role infeasibility during search, pruning illegal assignments early rather than filtering them after the fact:

$$\text{ss}_{e_1 e_2} \Rightarrow Z_{e_2, \text{lec}(e_1)} = 0 \quad \forall e_1, e_2 \in \mathcal{E} \quad (7)$$

H5 – No Double Invigilation. An instructor may invigilate at most one exam per timeslot:

$$\text{ss}_{e_1 e_2} \Rightarrow Z_{e_1 i} + Z_{e_2 i} \leq 1 \quad \forall i \in \mathcal{I} \quad (8)$$

H6 – Minimum Invigilator Count. Each exam must be assigned exactly $\text{req}^*(e) = \min(\text{req}(e), |\mathcal{I}|)$ invigilators:

$$\sum_{i \in \mathcal{I}} Z_{ei} = \text{req}^*(e) \quad \forall e \in \mathcal{E} \quad (9)$$

B. Soft Constraints and Objective Function

The system minimises:

$$\min F = w_1 S_1 + w_2 S_2 + w_3 S_3 + w_4 S_4 + S_5 + 5000 \cdot |\text{overflow}| \quad (10)$$

with default weights $w_1=1, w_2=5, w_3=2, w_4=3$.

S_1 – Time Preference. Penalises assigning instructors on declared leave days, implemented via `add_element` for variable-indexed array lookup.

S_2 – Workload Fairness (Min-Max Proxy). This objective improves upon prior CP formulations that either omit fairness or approximate it with variance. Variance minimisation is quadratic and unsupported natively in CP-SAT; moreover, a low variance is compatible with a large absolute workload gap. The min-max proxy:

$$S_2 = \max_{i \in \mathcal{I}} (\text{load}_i) - \min_{i \in \mathcal{I}} (\text{load}_i), \quad \text{load}_i = \sum_e Z_{ei} \quad (11)$$

directly encodes the worst-case overload as the optimisation target. CP-SAT supports `add_max_equality` and `add_min_equality` natively, so S_2 is linearised without auxiliary approximation. Minimising S_2 to zero guarantees perfect load equality; minimising variance to zero only guarantees equal second moments.

S_3 – Consecutive Invigilation. Penalises back-to-back assignments for the same instructor on the same day, implemented via $O(|\mathcal{I}| \times |\Gamma| \times |\mathcal{E}|)$ nested reification where Γ is the set of adjacent same-day timeslot pairs.

S_4 – Student Day Gap. Penalises student exam pairs with $|\text{day}(e_1) - \text{day}(e_2)| \leq 1$.

S_5 – Anti-Fragmentation. Minimises $\sum_{e,r} Y_{er}$ for physical rooms, discouraging unnecessary room splitting.

V. SYSTEM ARCHITECTURE AND IMPLEMENTATION

A. Data Parsing and Ingestion

A custom Python parser extracts exam, student, and instructor hierarchies from administrative spreadsheets using fuzzy string matching via Levenshtein distance to reconcile disparate spellings of instructor names and course codes, including Turkish character normalisation and prefix-based patterns. A 94.4% match rate was achieved on the University X dataset.

B. CP-SAT Solver Integration and Novelty of Encoding

Hard constraints are encoded as mandatory clauses via `model.Add()`; soft constraints contribute penalty terms in `model.Minimize()`. Three design choices distinguish our CP-SAT encoding from prior constraint-based formulations:

- 1) **Reified H4 clauses.** Dual-role conflict is encoded as a reified implication inside the solver, allowing CP-SAT’s propagator to infer timeslot infeasibility for a given instructor from the joint state of X_e and Z_{ei} variables, pruning the search tree far earlier than post-hoc filtering.
- 2) **Simultaneous Y_{er} - X_e propagation.** Because room and timeslot variables share the same constraint model, the solver can propagate room-capacity infeasibility (H2) to eliminate timeslot candidates (H1) in a single pass, an interaction impossible when room assignment is handled separately.
- 3) **Native min-max linearisation.** S_2 is encoded via `add_max_equality` and `add_min_equality` without any quadratic relaxation, preserving exact CP-SAT branch-and-bound optimality proofs.

The solver integrates AC-3 constraint propagation, Minimum Remaining Values (MRV) branching, Large Neighbourhood Search (LNS), and branch-and-bound for optimality proofs. All experiments used CP-SAT default seed (0) and OR-Tools version 9.x.

C. Asymptotic Complexity Analysis

Table II summarises variable and constraint counts per model component. The dominant memory bottleneck is H3 at $O(|\mathcal{E}|^2 \times |\mathcal{R}|)$, approximately 480,000 exclusion constraints for the University X instance and a projected 8.1 million for the uta-s-92 instance. The S_3 term contributes the largest auxiliary variable count via nested reification and is treated as optional for large instances.

VI. EXPERIMENTAL SETUP AND RESULTS

The framework was evaluated on: (A) four instances from the Carter/Toronto benchmark suite and (B) a real-world dataset from University X (Istanbul Okan University, Faculty of Engineering and Natural Sciences, Fall 2024–2025). All

TABLE II
ASYMPTOTIC COMPLEXITY OF MODEL COMPONENTS

Component	Variables	Constraints
Timeslot (X)	$O(\mathcal{E})$	—
Room (Y)	$O(\mathcal{E} \times \mathcal{R})$	$O(\mathcal{E})$
Invigilator (Z)	$O(\mathcal{E} \times \mathcal{I})$	$O(\mathcal{E})$
H1 (Conflict)	—	$O(\mathcal{C})$
H3 (Room Clash)	$O(\mathcal{E} ^2)$ reified	$O(\mathcal{E} ^2 \times \mathcal{R})$
H4 & H5	Shared w/ Z	$O(\mathcal{E} ^2 \times \mathcal{I})$
S_1 (Preference)	$O(\mathcal{E} \times \mathcal{I})$	$O(\mathcal{E} \times \mathcal{I})$
S_3 (Consec.)	$O(\mathcal{I} \times \mathcal{T} \times \mathcal{E})$	$O(\mathcal{I} \times \mathcal{T} \times \mathcal{E})$

experiments used weights $w_1=1$, $w_2=5$, $w_3=2$, $w_4=3$, CP-SAT default seed (0), S_3 disabled, S_4 enabled, and a 600-second wall-clock limit. Carter instances used 40 synthetic instructors, 20 synthetic physical rooms, and 45 synthetic timeslots. Table III summarises instance characteristics.

TABLE III
DATASET CHARACTERISTICS AND CONFLICT METRICS

Param.	hec	sta	yor	ear	Okan
$ \mathcal{E} $	80	138	180	189	131
$ \mathcal{S} $	2823	549	919	1108	1190
Conflict %	42.8	19.3	30.6	27.3	17.9
$ \mathcal{T} $	45 ^a	45 ^a	45 ^a	45 ^a	62
$ \mathcal{R} $	20 ^a	20 ^a	20 ^a	20 ^a	29
$ \mathcal{I} $	40 ^a	40 ^a	40 ^a	40 ^a	36

^aSynthetically generated.

A. Carter Benchmark Results

All four instances satisfied every hard constraint (H1–H6) with zero overflow. The fifth instance, uta-s-92 (638 exams), was killed by the OS due to memory exhaustion. Complete results are in Table IV.

TABLE IV
CARTER BENCHMARK RESULTS (S_3 DISABLED, 600 s). S_2 GAP REPORTED AS (MAX LOAD – MIN LOAD); E.G., (9–9) DENOTES MAX = MIN = 9, GAP = 0. ALL FEASIBLE RESULTS ARE HIGH-QUALITY SOLUTIONS; OPTIMALITY GAPS ARE NOT FORMALLY BOUNDED FOR THESE RUNS (SEE LIMITATION 3 IN SECTION VII).

Instance	Config.	Status	F	$S1$	$S2$ gap	Time
hec-s-92 [†]	S_1+S_2	OPTIMAL	0	0	0 (9–9)	14.3 s
hec-s-92	$S_1+S_2+S_4$	FEASIBLE	2292	0	2 (7–5)	600 s
sta-f-83	$S_1+S_2+S_4$	FEASIBLE	5745	0	3 (6–3)	600 s
yor-f-83	$S_1+S_2+S_4$	FEASIBLE	5660	1	2 (6–4)	600 s
ear-f-83	$S_1+S_2+S_4$	FEASIBLE	7458	0	21 (22–1) [‡]	600 s
uta-s-92	$S_1+S_2+S_4$	OOM	—	—	—	killed

[†] S_1+S_2 only; full $S_1+S_2+S_4$ run (row 2) yields $F=2,292$.

[‡]Structural saturation: all 45 slots and 20 rooms occupied; see Section VI-A.

hec-s-92 (80 exams). In the S_1+S_2 -only configuration, the solver achieved a provably OPTIMAL solution in 14.3 s with fairness gap of 0 (all 40 instructors assigned exactly 9 invigilation duties each). In the full $S_1+S_2+S_4$ configuration, $F=2,292$, fairness gap of 2 (loads 5–7), zero preference violations, and only 35/45 timeslots used.

sta-f-83 (138 exams). With 10.4 exams per student on average, the highest enrollment ratio in the suite, the solver achieved $F=5,745$ with fairness gap of 3 and zero preference violations.

yor-f-83 (180 exams). This instance saturated all 45 timeslots and all 20 rooms. Despite full saturation, the fairness gap was only 2 with just 1 preference violation.

ear-f-83 (189 exams) – Fairness Stress Case. The fairness gap collapsed to 21 (loads 1–22). Three structural factors interact to produce this outcome. First, simultaneous timeslot saturation (all 45 slots filled) and room saturation (all 20 rooms fully occupied) eliminate the solver’s ability to redistribute load by shifting exams to alternative slots. Second, the high conflict density (27.3%) creates hotspot clusters in the conflict graph: approximately 38% of examinations share conflicts with a subset of six or fewer synthetic instructors, concentrating assignment pressure on a small fraction of the invigilator pool. Third, the S_4 penalty pool dominates the objective at this scale ($F=7,458$ vs. the $2 \times w_2 \times S_2=210$ fairness contribution), causing the solver to trade fairness for student day-gap improvements within the fixed time budget. This is a **structural infeasibility** rather than a solver failure: under simultaneous slot and room saturation the min-max proxy loses the degrees of freedom required to balance loads.

uta-s-92 (638 exams) – Scalability Boundary. The $O(|\mathcal{E}|^2 \times |\mathcal{R}|)$ H3 constraint explosion (≈ 8.1 M exclusion constraints) exceeded practical memory limits. Column generation decomposition is required for instances beyond approximately 300 exams.

B. Positioning Relative to Prior Methods

The Carter proximity cost metric evaluates only student exam spacing; our CSOP optimises a strictly richer problem formulation. Table V positions this work against representative prior methods. Note that N/A for proximity cost reflects an incommensurable objective rather than an inferior one: a direct comparison would require removing H4, H5, H6, and S_2 from our formulation (see Section VII).

TABLE V
COMPARISON WITH REPRESENTATIVE PRIOR METHODS

Method	Type	Prox.	Fair.	H4
Carter et al. [1]	Graph col.	~ 10.83	N/R	×
Burke & Newall [2]	Tabu	~ 10.10	N/R	×
Tung et al. [10]	GA	N/R	N/R	×
Mandal et al. [5]	GD	Comp.	N/R	×
This work	Exact CSOP	N/A*	0–3	✓

N/R = not reported. *See Section VII.

C. Real-World Application: University X Dataset

The model was applied to real Fall 2024–2025 exam data from University X: 131 examinations, 36 instructors (25 PhD dual-role), 29 rooms, and 62 timeslots across 8 examination days. Table VI gives results for three configurations.

The best configuration (S_3 off, 600 s) achieved: zero preference violations; fairness gap of 2 (loads 4–6, mean 5.0, total

TABLE VI
UNIVERSITY X RESULTS (131 EXAMS, 36 INSTRUCTORS). BEST
CONFIGURATION IN BOLD.

Config.	$S1$	$S2$ gap	$S4$	Overfl.	Obj. F	Time
S_3 off, 300 s	0	2 (6-4)	1631	0	3272	300 s
S_3 off, 600 s	0	2 (6-4)	1310	0	2630	600 s
S_3 on, 600 s	20	9 (10-1)	1877	0	5898	600 s

F for S_3 -off runs computed as $w_1S_1 + w_2S_2 + w_4S_4$
($S_3=0$, S_5 not reported separately); exact values may differ slightly.

180 assignments); zero overflow; and zero dual-role violations (H4) across all 25 PhD candidates. Extending solve time from 300 s to 600 s improved S_4 by 19.7% and reduced F by 19.7%, demonstrating that the solver continues to find improvements within the time budget.

Multi-room assignment. A key demonstration of Contribution 2 is the successful scheduling of a 446-student examination across 14 physical classrooms simultaneously. The solver was required to jointly assign a timeslot and a set of rooms whose combined capacity met enrollment, propagating capacity constraints (H2) and room clash constraints (H3) in a single pass. This co-optimisation is impossible when room assignment is handled as a post-processing step, and directly validates the Y_{er} matrix formulation as a practical, scalable mechanism for large-enrollment examinations.

D. Ablation Study: The Cost of S_3

Table VII quantifies the tradeoff when S_3 is enabled within the 600-second budget. Enabling S_3 degraded every other objective simultaneously because it introduces approximately 236,000 auxiliary Boolean variables via nested reification, overwhelming the solver’s optimisation capacity. S_3 should remain disabled for instances exceeding ~ 130 exams unless solve-time budgets exceed approximately 30 minutes.

TABLE VII
ABLATION STUDY: S_3 ENABLED VS. DISABLED (UNIVERSITY X, 600 s)

Metric	S_3 off	S_3 on	Change
S_1 (pref. viol.)	0	20	+20
S_2 (fair. gap)	2	9	+7 (+350%)
S_4 (day pen.)	1310	1877	+567 (+43%)
Objective F	2630	5898	+124%

Weights: $w_1=1$, $w_2=5$, $w_3=2$, $w_4=3$ (both runs).

VII. DISCUSSION AND LIMITATIONS

Exact vs. heuristic methods. The CP-SAT solver proves optimality on *hec-s-92* under the S_1+S_2 -only configuration in 14.3 s—a guarantee no metaheuristic can provide *for the same objective function*. It is important to note that this optimality proof applies to our specific multi-objective formulation (minimising S_1+S_2), not to the Carter proximity cost metric, which is a different and incommensurable objective. For larger instances, high-quality FEASIBLE solutions are returned within practical time limits, though optimality gaps for these runs are not yet formally bounded (see Limitation 3 below).

Novel CP encoding vs. prior CP work. Three structural novelties are most visible in comparison with prior formulations. First, H4 introduces a bipartite dependency between the X_e and Z_{ei} variable families absent from all reviewed CP models; the CP-SAT propagator exploits this dependency during branch-and-bound in ways unavailable to post-hoc filtering. Second, the joint Y_{er} - X_e network enables propagation flows that are impossible when room assignment is a separate post-processing module. Third, the native min-max encoding of S_2 preserves exact optimality certificates that a quadratic surrogate would forfeit.

The ear-f-83 boundary condition. The fairness collapse (gap 21 vs. 2–3 on other instances) is a structural finding: the min-max proxy works well when scheduling degrees of freedom are available but degrades under simultaneous timeslot and room saturation. Institutions near capacity should expand the instructor pool or increase the number of available timeslots as primary remedies. Adjusting w_2 relative to w_4 may partially compensate, but the extent of this effect has not been formally evaluated; a sensitivity analysis across weight configurations is identified as future work.

Proximity cost vs. multi-objective fairness. Comparing our model on the Carter proximity cost metric would require removing H4, H5, H6, and S_2 , reducing the formulation to standard graph colouring. The correct comparison is a richer constraint set vs. a simpler one, as Table V reflects. We report N/A to be precise rather than misleading: our objective function is incommensurable with the Carter metric, not simply a different approximation of it.

Scalability boundary. Column generation decomposition, separating timeslot assignment from room/invigilator routing, is required for instances beyond ~ 300 exams.

Limitations. (1) Invigilators are assigned to examinations as a whole rather than to specific rooms within a multi-room cluster; operational completeness requires invigilator-room assignment within clusters. (2) The S_3 formulation creates a variable explosion at scale ($\sim 236K$ auxiliary Booleans for the University X instance), making it impractical within a 600-second budget for instances exceeding ~ 130 examinations. (3) **All results beyond the *hec-s-92* S_1+S_2 run are FEASIBLE; optimality gaps are not yet reported.** For the University X real-world deployment in particular, the solution quality relative to the true optimum is unknown. Reporting lower bounds or optimality gaps for at least one additional instance is identified as a priority for any subsequent journal submission. (4) The ear-f-83 fairness collapse under saturation has not been resolved algorithmically.

Reproducibility. All experiments used $w_1=1$, $w_2=5$, $w_3=2$, $w_4=3$, CP-SAT default seed (0), OR-Tools v9.x, S_3 disabled, S_4 enabled, 600 s time limit. Carter instances (version 2) were sourced from the Nottingham benchmark repository [3]. Solver code will be released upon acceptance.

VIII. CONCLUSION AND FUTURE WORK

This paper presented a CSOP framework for university examination timetabling that advances the state of CP-based

approaches along three previously unaddressed dimensions: formally encoding dual-role PhD instructor conflicts as a hard constraint (H4), integrating multi-room Boolean assignment as a first-class CP decision variable, and optimising invigilator workload fairness via a min-max proxy that is both natively supported by CP-SAT and theoretically stronger than variance minimisation.

Validation on four Carter benchmark instances and one real-world institutional dataset yielded four key findings: optimal fairness on hec-s-92 (under the S_1+S_2 configuration); near-optimal fairness (gap 2–3) on sta-f-83 and yor-f-83; a structural fairness collapse under simultaneous saturation on ear-f-83; and a +124% objective increase when S_3 is enabled within a fixed time budget. The multi-room assignment capability was demonstrated concretely by scheduling a 446-student examination across 14 simultaneous physical classrooms.

Four directions for future work are identified: (1) S_3 reformulation using CP-SAT interval variables or `add_no_overlap` to reduce the ~236K auxiliary variable overhead; (2) column generation decomposing large instances into timeslot-assignment and room/invigilator-routing subproblems; (3) warm starting from existing manual schedules to reduce solve time by up to 70%; and (4) invigilator-room assignment within multi-room clusters for operational completeness.

ACKNOWLEDGMENT

The authors would like to express their sincere gratitude to Assistant Professor Sina Alp for his invaluable guidance, technical support, and insightful comments throughout the course of this research. His expertise and encouragement were instrumental in the successful completion of this work.

REFERENCES

- [1] M. W. Carter, G. Laporte, and S. Y. Lee, “Examination timetabling: Algorithmic strategies and applications,” *J. Oper. Res. Soc.*, vol. 47, no. 3, pp. 373–383, 1996.
- [2] E. K. Burke and J. P. Newall, “Solving examination timetabling problems through adaption of heuristic orderings,” *Ann. Oper. Res.*, vol. 129, pp. 107–134, 2004.
- [3] R. Qu, E. K. Burke, B. McCollum, L. T. G. Merlot, and S. Y. Lee, “A survey of search methodologies and automated system development for examination timetabling,” *J. Scheduling*, vol. 12, no. 1, pp. 55–89, 2009.
- [4] S. Ceschia, L. Di Gaspero, and A. Schaerf, “Educational timetabling: Problems, benchmarks, and state-of-the-art results,” *arXiv preprint arXiv:2201.07525*, 2022.
- [5] A. K. Mandal, M. N. M. Kahar, and G. Kendall, “Addressing examination timetabling problem using a partial exams approach in constructive and improvement,” *Computation*, vol. 8, no. 2, p. 46, 2020.
- [6] S. Bazari, A. Pooya, O. S. Fard, and P. Roozkhosh, “Modeling and solving the problem of scheduling university exams in terms of new constraints on professors’ exam conflicts,” *OPSEARCH*, vol. 60, pp. 877–915, 2023.
- [7] M. Comlan and C. Allohoumbo, “Constraint satisfaction algorithms: Edition of timetables in the license-master-doctorate system,” *Comput. Sci. Inf. Technol.*, vol. 4, no. 3, pp. 217–225, 2023.
- [8] H. Gültekin, S. Z. Qamar, and H. Al-Lawati, “A practical algorithm for midterm exam scheduling in higher education institutions,” in *Proc. 14th Ann. Int. Conf. Ind. Eng. Oper. Manage.*, Dubai, UAE, 2024.
- [9] T. Alrawashdeh, K. G. Al-Moghrabi, and A. M. Al-Ghonmein, “A profiling-based algorithm for exams’ scheduling problem,” *Int. J. Electr. Comput. Eng.*, vol. 13, no. 5, pp. 5483–5490, 2023.
- [10] S. N. Tung, J. B. Jaafar, I. A. Aziz, H. G. Nguyen, and A. N. Bui, “Genetic algorithm for solving multi-objective optimization in examination timetabling problem,” *Int. J. Emerg. Technol. Learn.*, vol. 16, no. 11, 2021.
- [11] A. Ghaffar, I. U. Din, M. H. Zafar, and A. Tariq, “Hybridization and artificial intelligence in optimizing university examination timetabling problem: A systematic review,” *Rev. Educ.*, vol. 13, e70071, 2025.
- [12] A. Bonlarron, F. Regin, E. De Maria, and J.-C. Régin, “Large language model meets constraint propagation,” in *Proc. 34th Int. Joint Conf. Artif. Intell. (IJCAI-25)*, 2025.
- [13] L. Xiaofan, X. Jiaqian, and L. Xiaolong, “The exam timetable problem with examination conflicts,” in *Proc. 2023 6th Int. Conf. E-Bus. Inf. Manage. Comput. Sci.*, 2023.
- [14] M. Shakir et al., “Examination time table generator,” *Int. J. Sci. Res. Eng. Manage.*, vol. 09, no. 01, 2025.
- [15] S. D. Gupta, B. Genc, and B. O’Sullivan, “Explanation in constraint satisfaction: A survey,” in *Proc. 30th Int. Joint Conf. Artif. Intell. (IJCAI-21)*, 2021, pp. 4400–4407.
- [16] J. Khun and J. Schmidt, “Interval constraint satisfaction: Towards edge acceleration,” in *2022 11th Mediterranean Conf. Embedded Computing (MECO)*, 2022.
- [17] E. S. K. Siew, S. N. Sze, S. L. Goh, M. Hossin, and K. L. Chiew, “Multi-neighborhood local search with room split balancer for exam timetabling: A case study,” *Math. Model. Comput.*, vol. 12, no. 1, pp. 144–157, 2025.
- [18] T. Onagi, K. Mizuno, and K. Sato, “Effectiveness of an ACO algorithm with Lévy flight for large-scale constraint satisfaction problems,” in *Proc. 2024 IEEE Congr. Evol. Comput. (CEC)*, 2024.
- [19] S. Selimi, L. Arbnesi, K. Sylejmani, and N. Musliu, “Iterated local search for the examination timetabling problem with constructive-based initial solution,” in *Proc. 13th Int. Conf. Practice Theory Autom. Timetabling (PATAT 2022)*, vol. III, 2022.
- [20] J. Tönshoff, B. Kisin, J. Lindner, and M. Grohe, “One model, any CSP: Graph neural networks as fast global search heuristics for constraint satisfaction,” in *Proc. 32nd Int. Joint Conf. Artif. Intell. (IJCAI-23)*, 2023.